higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONICS
NQF LEVEL 2

(10041012)

27 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Round off ALL final answers to THREE decimal places.
5. Write neatly and legibly.

NCV

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Output is logic one if ALL inputs are logic zero:

A OR gate
B NAND gate
C NOR
D AND

1.1.2 According to Ohm's law, current is given by which ONE of the following equations:

A $I = P \times V$
B $I = \frac{R}{V}$
C $I = \frac{V}{R}$
D $I = \frac{V^2}{R}$

1.1.3 Which ONE of the following is NOT an effect of electrical shock:

A Memory loss
B Tightening of muscles
C Internal bleeding
D Breathing difficulties

1.1.4 Which ONE of the following tools is NOT used during soldering:

A Soldering iron
B Pliers
C Drilling machine
D Helping hands

1.1.5 Which ONE of the following must be observed when measuring voltage with an analogue meter:

A Observe polarity, connect multimeter in series, select ammeter reading/scale
B Observe polarity, connect multimeter in parallel, select voltage reading/scale, select the highest scale
C Observe polarity, connect multimeter in series, select voltmeter reading/scale
D NO need to observe polarity, connect multimeter in parallel, select ammeter reading/scale

(5 × 1) (5)

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1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A permanent magnet loses its magnetic strength if the electric field is removed.

1.2.2 SANS 10142 describes how to perform safe electrical installations and repairs.

1.2.3 10001_2 converted to decimal is 9_{10}.

1.2.4 A P-type has more electrons than holes.

1.2.5 To avoid electrical shock or damage to a meter, do NOT exceed input limits.

(5 × 1) (5)

1.3 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Energy converted from radiant energy into electricity

1.3.2 Formed when pentavalent impurity atoms are added to silicon material

1.3.3 Light that shows that a PLC program or system is having errors

1.3.4 Where there are slots to plug in input and output devices

1.3.5 Computers that cover a wider area

(5 × 1) (5)
[15]

TOTAL SECTION A: 15

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SECTION B

QUESTION 2

2.1 Explain how electricity is generated from each of the following sources:

2.1.1 Wind

2.1.2 Nuclear

(2 × 2) (4)

2.2 Explain how current relates to voltage and resistance according to Ohm's law. (2)

2.3 Calculate the current flowing through the conductor in FIGURE 1 if the following information is given:

Length of the conductor (L): 12 m
Number of conductors or turns (N): 800
Magnetic field strength (H): 50 Am^{-1} (3)

2.4

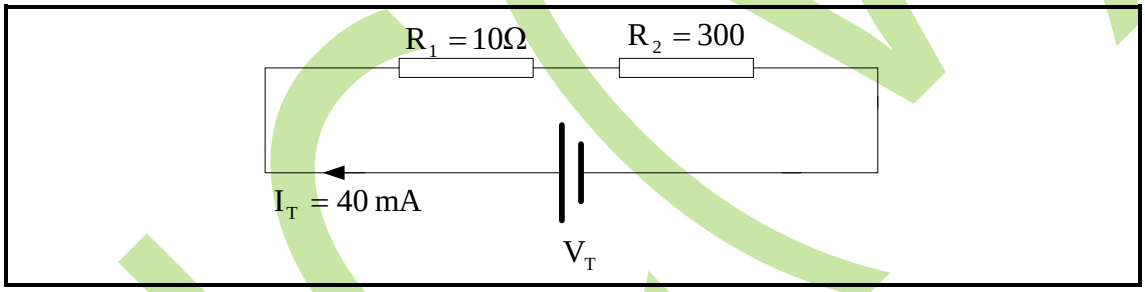


FIGURE 1

Use the circuit in FIGURE 1 to calculate the following:

2.4.1 Total resistance of the circuit

2.4.2 Supply voltage V_T

2.4.3 Power dissipated by 300 Ω

(3 × 2) (6)

2.5 What is the colour coding for a 3 300 ±5% Ω resistor? (4)

2.6 Name each of the following IEC circuit symbols:

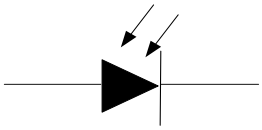
2.6.1



2.6.2



2.6.3



(3 × 1) (3)

2.7 Refer to FIGURE 2 and assume that the diode is made up from silicon material (voltage across the diode V_D = 0,7 V).

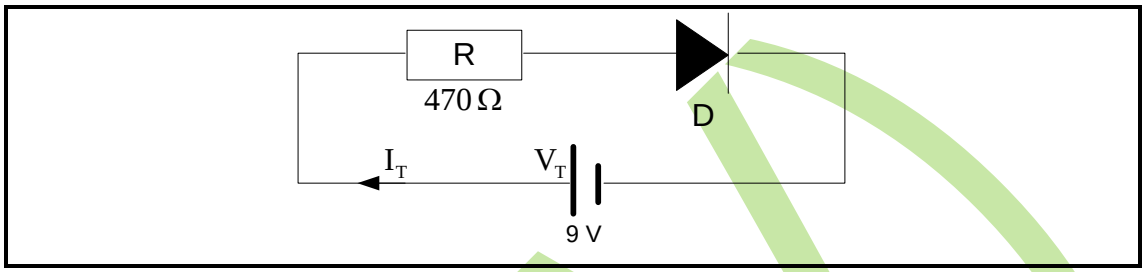


FIGURE 2

Calculate the following:

2.7.1 Voltage across the resistor (3)

2.7.2 Current flowing through the circuit (2)

2.8

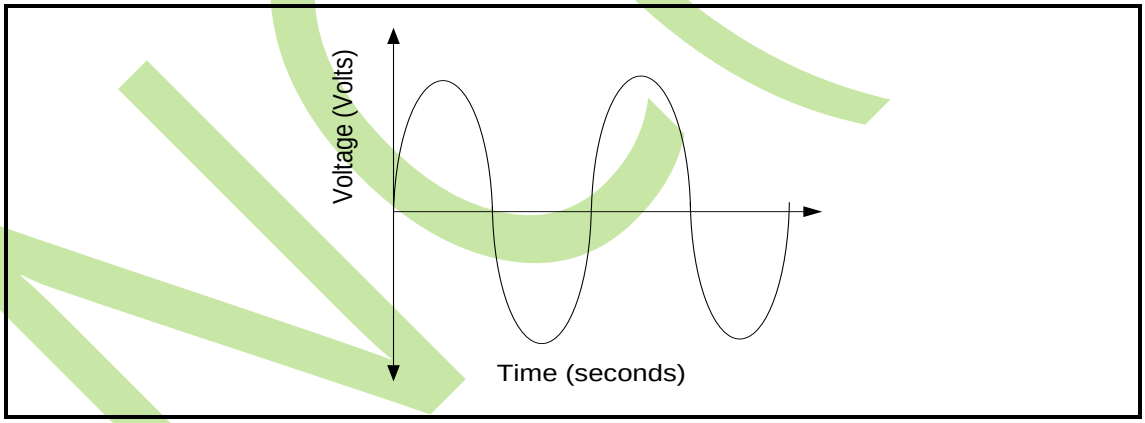


FIGURE 3

Study FIGURE 3 and draw the output waveforms if the given waveform is the output of a transformer supplying a:

2.8.1 Half-wave rectifier

2.8.2 Full-bridge rectifier

2.8.3 Centre-tap transformer rectifier

(3 × 1) (3)
[30]

QUESTION 3

3.1 What is the importance of earthing in an electrical installation? (5)

3.2 Technicians must always take precautions when measuring the current or voltage in a live circuit.

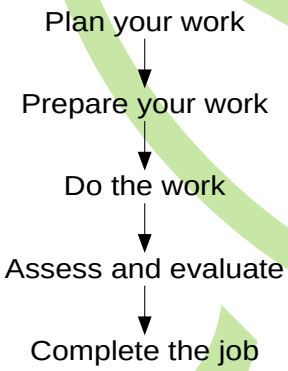
Which safety precautions must be taken when measuring either current or voltage with an analogue meter? (5)

3.3 The first step in soldering joints is tinning.

3.3.1 Explain the term *tinning*. (1)

3.3.2 Explain how tinning of components is done. (4)

3.4 The flow chart below outlines the steps to follow when planning any project or job.



Explain how to execute step 2. (5)
[20]

QUESTION 4

4.1 Convert each of the following binary numbers to the decimal system:

4.1.1 1101_2

4.1.2 11001_2

(2 × 2) (4)

4.2 Calculate each of the following in the binary system:

4.2.1 $1101_2 + 11001_2$

4.2.2 $11001_2 – 1101_2$

(2 × 2) (4)

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4.3 FIGURE 4 shows a switch circuit.

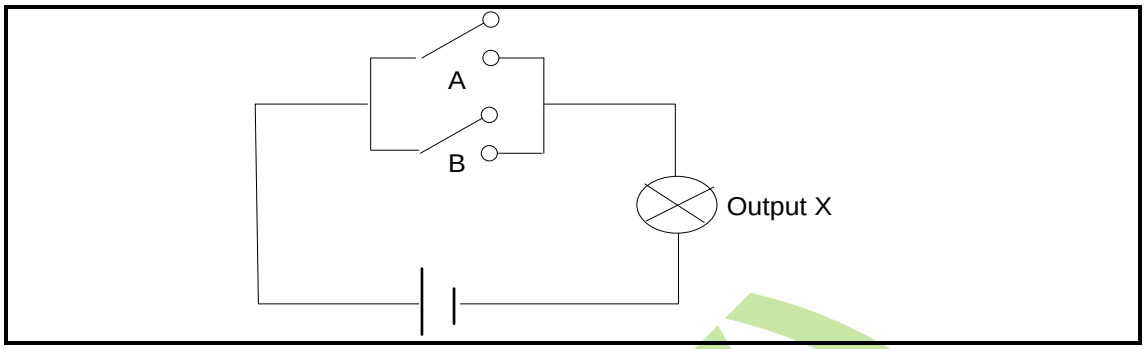


FIGURE 4

4.3.1 Name the logic gate shown in FIGURE 4. (1)

4.3.2 Draw the truth table for the logic gate in QUESTION 4.3.1. (4)

4.3.3 Write a mathematical equation that will give the same results as the answer in QUESTION 4.3.2. (1)

4.3.4 Draw the IEC symbol of the logic gate in QUESTION 4.3.1 if the output X is connected in parallel with the switches. (1)
[15]

QUESTION 5

5.1 Briefly explain how to establish communication between the *programmer* and *processor.* (5)

5.2 Explain the principle of *multiplexing* as it is used in telecommunications. (5)

5.3 Tabulate the difference between *amplitude modulation* and *frequency modulation*. (3 + 3) (6)

5.4 Name and describe TWO computer networks. (2 × 2) (4)
[20]

TOTAL SECTION B: 85
GRAND TOTAL: 100

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